llmopt + axiom | two measured research labs

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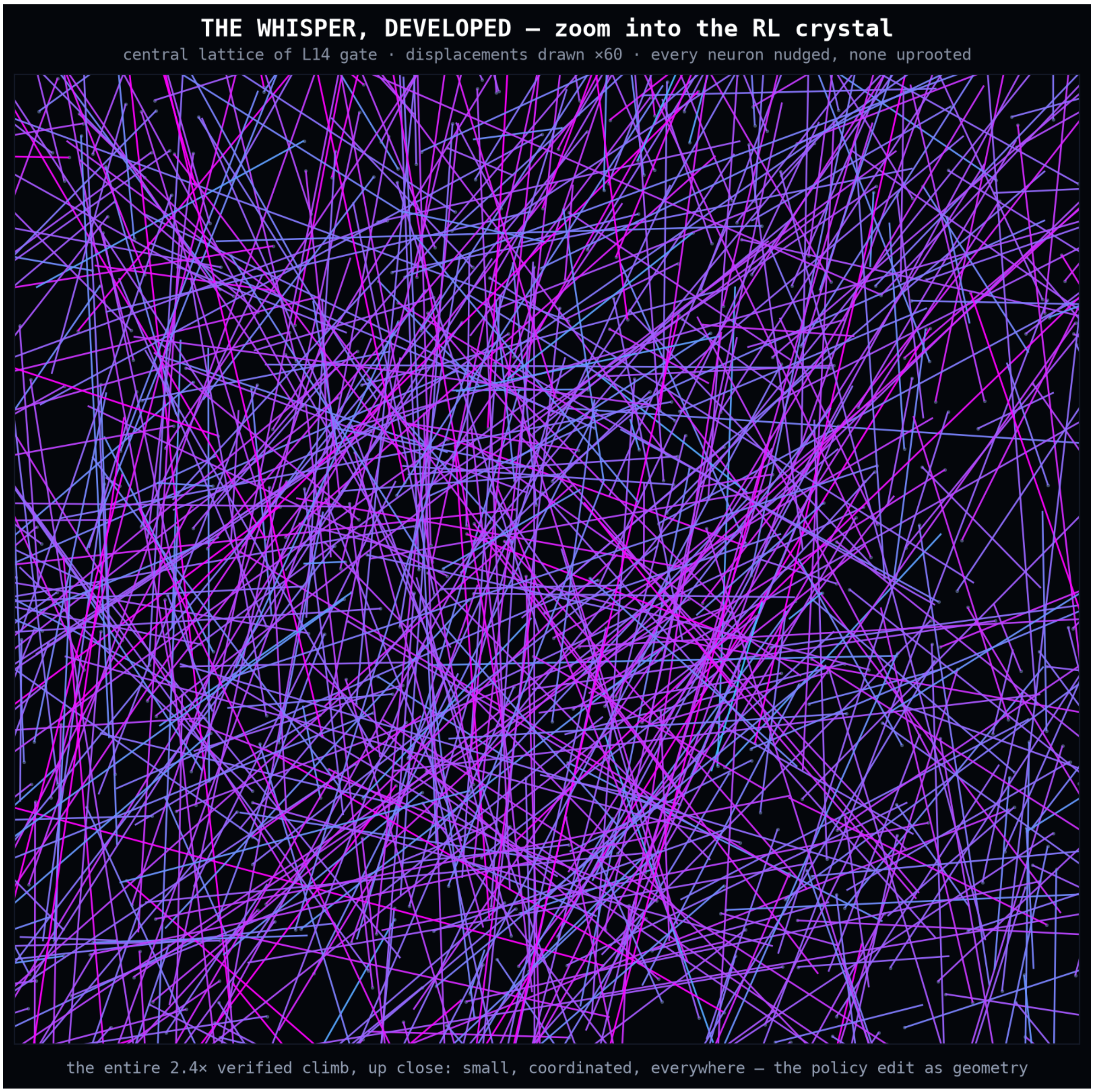
Open research lab for closed-system mathematical models and mixture-of-experts behavior, with chess-engine-style evaluation discipline: pre-registered experiments, paired seeds, oracle-verified scoring, and honest nulls kept in an append-only ledger.

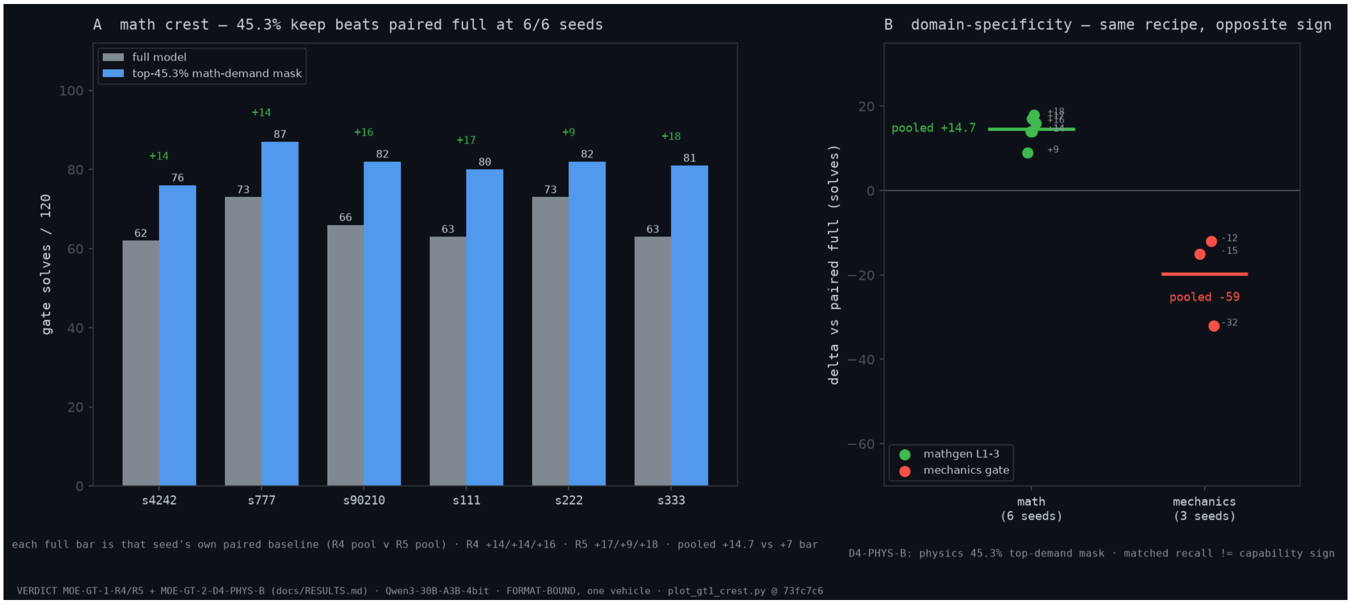
Paired with axiom, an independent C++ derivation engine: the two labs verify each other's artifacts, so "reproduced" means a second implementation agrees, not a rerun of the same code.



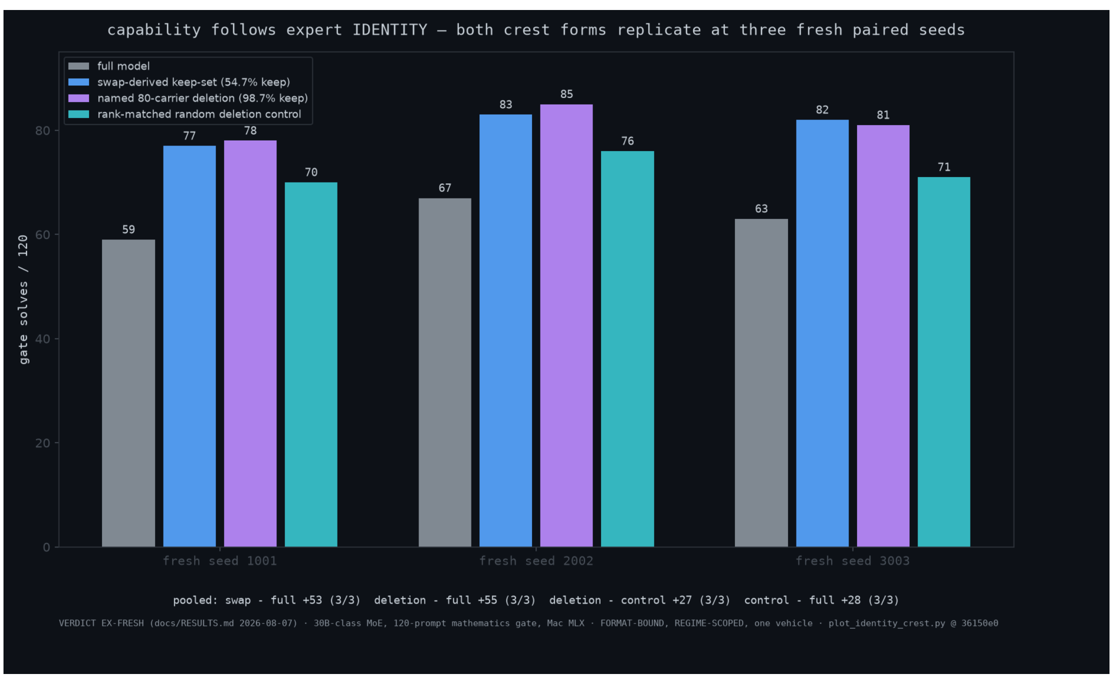
One layer of a 0.5B model during a verified 2.4x reinforcement-learning capability climb, each line one neuron's movement drawn at 60x magnification: every neuron nudged, none uprooted. The entire climb wrote ~6% of one supervised run's weight movement (delta-W 4.0 vs 61-87), with representations near-identical before and after (CKA 0.9998). RL edits the policy, not the representation.

Expert identity beats aggregate summaries

30B-class mixture-of-experts, 120-item oracle-scored gates



Keeping the top 45.3% of experts by math demand beats the full model on the math gate, 6/6 paired seeds (+14.7 pooled). The same recipe inverts on a mechanics gate (-59 pooled): domain-specific identity, not generic sparsity. Regenerates from a committed script; provenance footer in-image.



Decomposition: swap-derived keep-set (+53), deletion of 80 named interfering experts (+55), rank-matched control (+28), each 3/3 at fresh seeds. Capability follows WHICH experts are kept, not how many.

Two labs, one verification loop

Every headline number below is booked in the public ledger

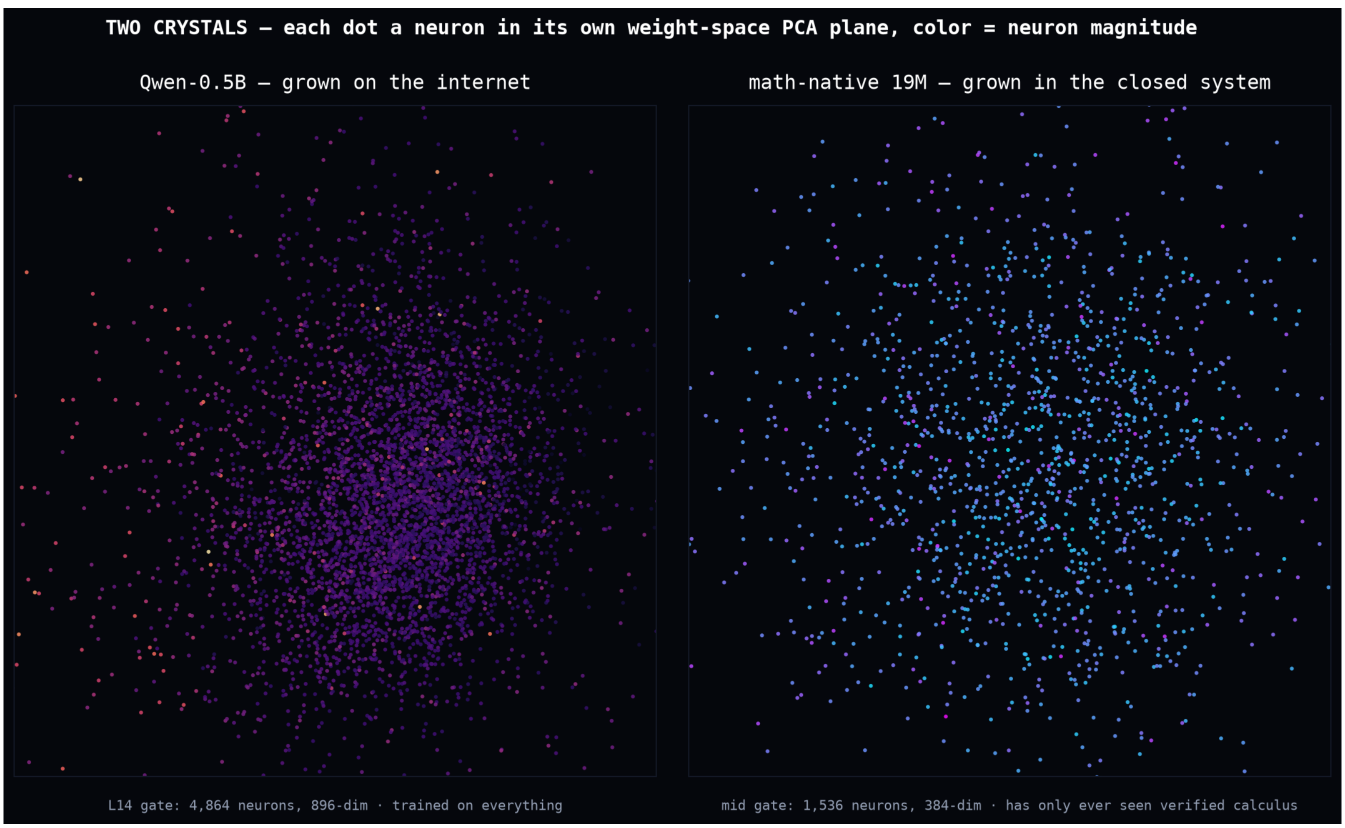
360/360 on held-out, sympy-oracle-verified calculus (up from 73.6%), every gain tied to a named component.

Cross-lab replay: 50/50 trajectories token-identical between llmopt's Python engine and axiom's C++ leg.

Certified row factory: 167/167 axiom-emitted training rows pass llmopt's independent oracle, schema-exact; emission audits 5-for-5 clean (latest: 0 contaminated rows in 145,011).

21,614/21,914 kernel-certified Lean 4 certificates (98.63%, closed failure taxonomy); neural-net interchange format (AXNN) at 20/20 cross-lab parity.

Exactness work: integer and deterministic compute paths with digest and trajectory checks, so agreement is bit-level where the instrument allows it.



Same lens, two diets: web-pretrained 0.5B (dense isotropic cloud) vs the closed-system 19M math native (sparse ring structure). You can see the training diet in the weights.

Philosophy: small, fully owned training paths beat un-auditable scale for causal claims; nulls and between-threshold outcomes are first-class results.